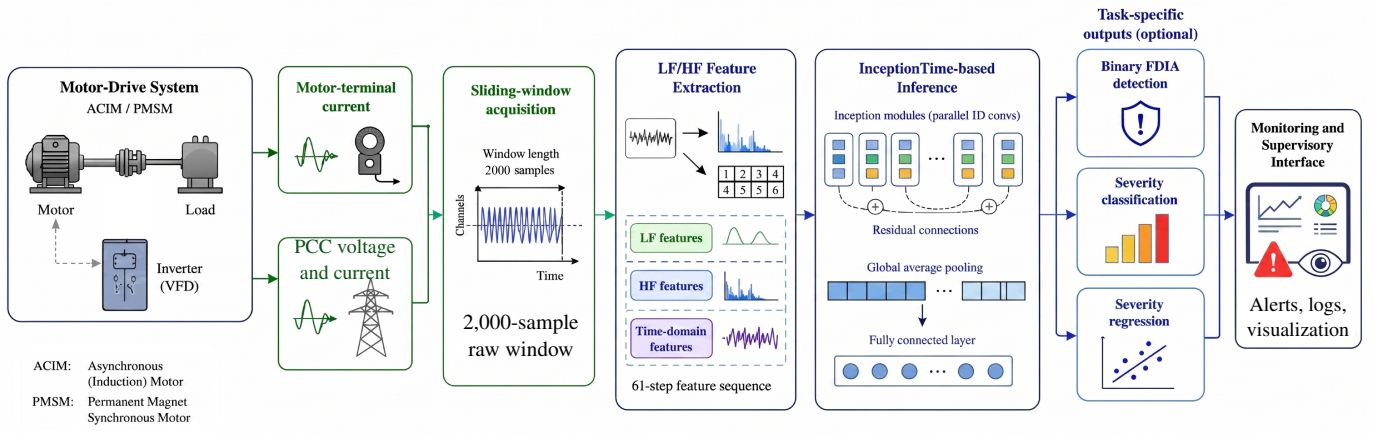

Motor-Drive System
ACIM / PMSM
Motor
Load
Inverter
(VFD)
ACIM: Asynchronous (Induction) Motor
PMSM: Permanent Magnet Synchronous Motor
Motor-terminal current
PCC voltage and current
Sliding-window acquisition
Window length
2000 samples
Channels
Time
2,000-sample raw window
LF/HF Feature Extraction
1 2 3 4
4 5 5 6
LF features
HF features
Time-domain features
61-step feature sequence
InceptionTime-based Inference
Inception modules (parallel ID convs)
Residual connections
Global average pooling
Fully connected layer
Task-specific outputs (optional)
Binary FDIA detection
Severity classification
Severity regression
Monitoring and Supervisory Interface
Alerts, logs, visualization